

RESEARCH PROJECT REPORT

Beyond the Tumor: Integrating Transcriptomic Dysregulation and Immune Signatures for Survival Prediction in Lung Adenocarcinoma

A Multi-Modal Machine Learning Approach

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Bioinformatics Lead: Sonica
Datasets: TCGA-LUAD · GTEx · GEO · CPTAC-LUAD
Report covers: Phases 3 & 4 (Clinical EDA + ML Models)
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484	478	1,000	0.712
Patients cleaned	Matched patients	Genes selected	DeepSurv C-index

This document covers the ML pipeline implementation from raw TCGA data to trained deep learning survival models. All code is available on GitHub. Results are fully reproducible.

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1 Project Overview

1.1 Scientific Question

Core Research Question

Can patient-specific tumour-normal transcriptomic dysregulation, combined with immune microenvironment signatures, improve survival prediction in Lung Adenocarcinoma compared to models using tumour gene expression alone?

Lung adenocarcinoma (LUAD) is the most common subtype of non-small cell lung cancer and remains one of the leading causes of cancer-related death worldwide. Despite improvements in targeted therapies and immunotherapy, patient outcomes remain highly variable — two patients with the same stage and treatment can have dramatically different survival outcomes.

Most current survival prediction models look at tumour gene expression in isolation. This project proposes to combine three streams of biological information: tumour gene expression, tumour-normal transcriptomic dysregulation, and immune microenvironment profiles into a single multi-modal machine learning model to predict survival in LUAD patients.

1.2 Task Division

Task	Sonica	Parth (ML)
Data download & cleaning	Primary	Support
GTEX normalisation	Primary	—
Dysregulation z-score computation	Primary	—
Immune deconvolution (CIBERSORT)	Primary	—
Clinical feature curation	Primary	—
Cox-Lasso baseline models	Primary	Support
DeepSurv implementation	—	Primary
Multi-modal fusion architecture	—	Primary
SHAP analysis	Support	Primary
Biological interpretation of SHAP	Primary	Support

Table 1. Task division between team members.

1.3 Pipeline Overview

The full ML pipeline implemented in this report follows this sequence:

Step 1. Clinical data loading and cleaning — Extract survival endpoints from TCGA-LUAD

Step 2. Exploratory data analysis — Kaplan-Meier curves, distribution analysis

Step 3. Cox baseline model — Age, stage, gender as features

Step 4. RNA-seq processing — Load, clean, normalise, match patients

Step 5. Gene selection — Two-stage: variance filter + univariate Cox

Step 6. Cox-Lasso on expression — Linear survival model on 1,000 genes

Step 7. DeepSurv — Neural network with Cox partial likelihood loss

2 Data Sources and Preprocessing

2.1 Clinical Data

Property	Detail
Source	GDAC Broad Institute (Firehose, 2016-01-28 release)
File	LUAD.merged_only_clinical_clin_format.txt
Raw dimensions	527 clinical variables $\times$ 522 patients
Format	Tab-separated, transposed (rows=variables, columns=patients)
Missing values	Encoded as NA, [Not Available], [Unknown]
Patient IDs	TCGA barcodes: tcga-05-4244

Table 2. Clinical data source details.

2.1.1 Survival Column Construction

Six columns were extracted from 527 clinical variables:

Column	Renamed To	Role
patient.vital_status	vital_status	Dead or alive at last contact
patient.days_to_death	days_to_death	Days to death (dead patients only)
patient.days_to_last_followup	days_to_last_followup	Days tracked (censored patients)
patient.age_at_initial_pathologic_diagnosis	age	Age at diagnosis (years)
patient.gender	gender	Biological sex
patient.stage_event.pathologic_stage	stage	TNM pathologic stage

Table 3. Clinical columns extracted and renamed.

Two derived columns were constructed:

- **survival_time** = `days_to_death` if dead, else `days_to_last_followup`
- **event** = 1 if `vital_status == dead`, else 0

2.1.2 Data Cleaning

Step	Removed	Reason	Remaining
Raw load	—	Starting point	522
Drop missing survival.time	19	No survival information	503
Drop zero survival.time	19	Biologically implausible	484
Final dataset	38 total		484

Table 4. Clinical data cleaning steps. Total data loss: 7.3%.

2.2 RNA-seq Data

Property	Detail
Source	GDAC Broad Institute, RSEM genes normalised
File	LUAD.rnaseqv2...RSEM_genes_normalized_data.data.txt
Raw dimensions	20,531 genes $\times$ 576 patients
Normalisation applied	$\log_2(x + 1)$ on RSEM values
Patient ID format	TCGA-05-4244-01A-01R-1107-07 (trimmed to 12 chars)

Table 5. RNA-seq data source details.

2.2.1 RNA-seq Cleaning Pipeline

- Skip junk header row** — File contains two header rows; second row (`normalized_count`) skipped via `skiprows=[1]`
- Remove unnamed genes** — 29 genes with ? symbol removed; 20,531 $\rightarrow$ 20,502 genes
- Clean gene names** — `EGFR|1956` $\rightarrow$ `EGFR`
- Trim patient IDs** — First 12 characters, lowercased to match clinical
- Remove duplicate patients** — 60 patients with two tumour samples; kept first occurrence; 576 $\rightarrow$ 516 unique patients
- Patient matching** — Inner join on patient IDs; $484 \text{ clinical} \cap 516 \text{ RNA-seq} = \mathbf{478}$ matched patients
- Normalisation** — Applied $\log_2(x + 1)$; range compressed from $[0, 1,432,694]$ to $[0, 20.45]$

2.3 Gene Selection — Two-Stage Filter

With 20,502 genes and only 478 patients, dimensionality reduction is essential to prevent overfitting. We applied a two-stage selection strategy:

Stage	Method	Rationale	Genes
0	Raw (after cleaning)	Starting point	20,502
1	Variance filter	Remove completely flat, uninformative genes	5,000
2	Univariate Cox	Keep genes most associated with survival	1,000
Final gene matrix			1,000

Table 6. Two-stage gene selection pipeline.

Why Two Stages?

Variance alone does not guarantee survival relevance — a gene can vary due to technical noise rather than biology. Stage 1 cheaply removes flat genes. Stage 2 directly connects selection to the prediction target by fitting a univariate Cox model per gene and retaining the 1,000 most significantly survival-associated genes (p-value cutoff: $p < 0.073$).

Top 10 survival-associated genes identified by univariate Cox:

Rank	Gene	p-value	Known Biology
1	LINGO2	1.51×10^{-7}	Leucine-rich repeat neuronal protein
2	EPGN	2.26×10^{-6}	Epidermal growth factor, cancer proliferation
3	DKK1	2.43×10^{-6}	Wnt pathway inhibitor, LUAD prognostic marker
4	CD109	3.52×10^{-6}	TGF- β regulator, poor prognosis marker
5	IRX5	1.44×10^{-5}	Transcription factor, cancer cell proliferation
6	DNAH2	2.63×10^{-5}	Dynein heavy chain
7	INPP5J	3.13×10^{-5}	Phosphatase, PI3K pathway
8	FGF5	3.89×10^{-5}	Fibroblast growth factor, tumour angiogenesis
9	NTSR1	4.22×10^{-5}	Neurotensin receptor, lung cancer biomarker
10	PCSK9	—	Cholesterol metabolism, cancer metabolism

Table 7. Top 10 genes most significantly associated with survival by univariate Cox regression. Appearance of known cancer biology genes (DKK1, CD109, FGF5) validates the selection approach.

3 Exploratory Data Analysis

3.1 Clinical Data Overview

Variable	Value	Notes
Total patients	484	After cleaning
Death events	122 (25.2%)	Uncensored observations
Censored patients	362 (74.8%)	Alive at last contact
Median survival time	248 days	~8 months
Mean survival time	543 days	~1.5 years
Survival range	1 – 6,812 days	Max ~18.7 years
Median age at diagnosis	66 years	Range: 35–89
Gender split	224M / 260F	46% male, 54% female

Table 8. Clinical cohort summary statistics.

3.2 EDA Figures

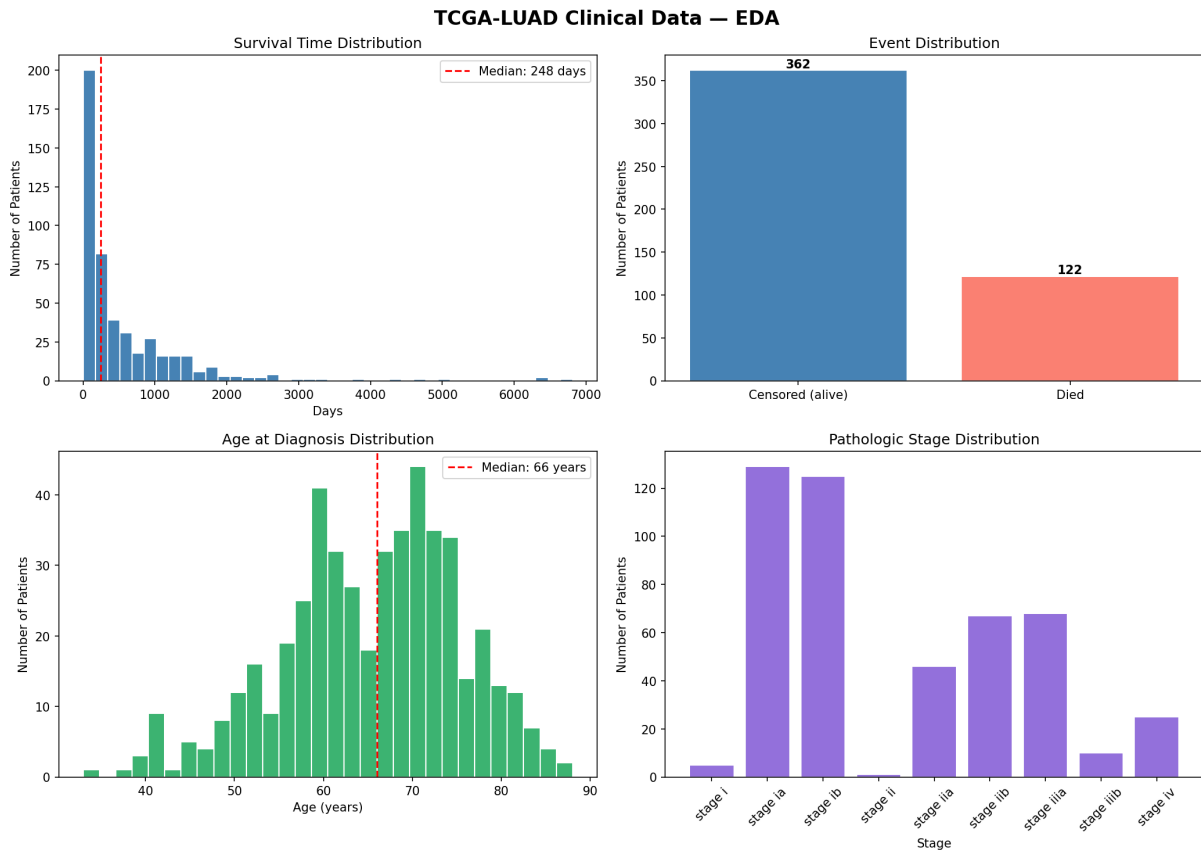


Figure 1. Clinical EDA — Four-panel summary. *Top left:* Survival time distribution showing strong right skew — the majority of patients cluster within 500 days with a long tail of long-term survivors (median = 248 days). This right-skewed, non-normal distribution justifies the use of survival analysis rather than linear regression. *Top right:* Event distribution showing 362 censored and 122 deceased patients (25.2% event rate). *Bottom left:* Age at diagnosis approximately bell-shaped around median 66 years, consistent with LUAD being predominantly a disease of older adults. *Bottom right:* Stage distribution showing over-representation of Stage Ia and Ib, reflecting TCGA’s sampling bias toward early-stage, surgically resectable tumours.

3.3 Kaplan-Meier Survival Curves

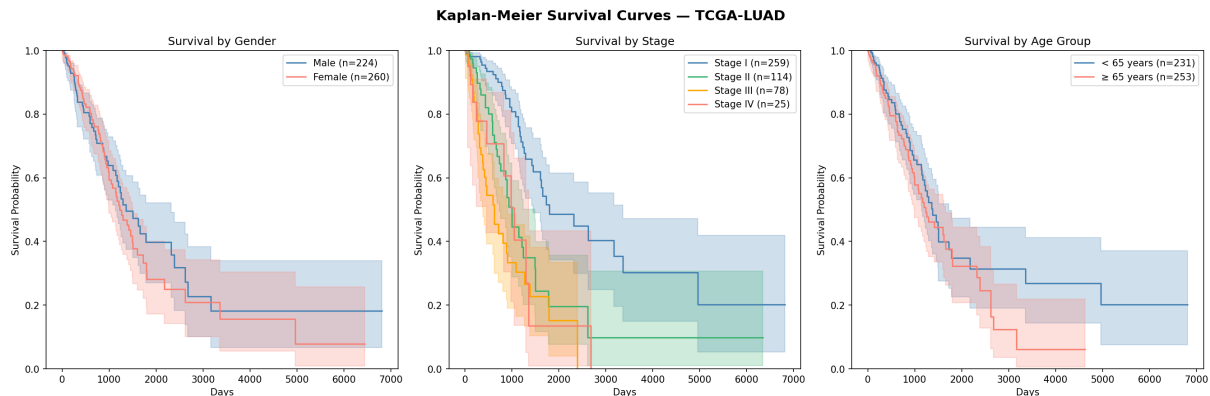


Figure 2. Kaplan-Meier survival curves by clinical subgroup. Each curve shows the probability of survival over time for a patient subgroup. Steps in the curve occur at death events; tick marks indicate censored patients. Shaded regions represent 95% confidence intervals. *Left:* By gender — male patients show slightly higher survival probability throughout follow-up, though overlapping confidence intervals indicate this difference is not statistically significant (confirmed by Cox $p=0.19$ for `is_male`). *Centre:* By pathologic stage — clear separation in expected order (Stage I > II > III > IV). Stage IV patients ($n=25$) show near-complete mortality by 1,000 days. This is the strongest visual predictor in the dataset. *Right:* By age group — younger patients (<65 years) consistently survive longer, with meaningful separation especially beyond 1,000 days.

Key Clinical Finding from Kaplan-Meier Analysis

Pathologic stage is the dominant clinical predictor of survival. The stage-stratified curves show clear, consistent separation without crossing — satisfying the proportional hazards assumption of the Cox model. Gender shows weak signal. Age shows moderate signal. These findings directly inform feature selection for the Cox baseline model.

4 Cox Proportional Hazards Baseline (Model 1)

4.1 Model Description

The Cox proportional hazards model estimates the effect of each variable on the instantaneous risk (hazard) of death. It outputs a hazard ratio (HR) per variable, interpretable as a risk multiplier relative to a reference group:

- $HR > 1.0$: variable increases risk of death
- $HR < 1.0$: variable decreases risk (protective)
- $HR = 1.0$: no effect on survival

Features used: age (continuous), is_male (binary, female=reference), stage_II, stage_III, stage_IV (binary dummies, Stage I=reference).

Implementation: `lifelines.CoxPHFitter`, Breslow baseline estimation.

4.2 Results

Variable	Coef	HR	95% CI	p-value	Significance
age	0.02	1.02	[1.00 – 1.04]	0.05	Borderline
is_male	-0.25	0.78	[0.54 – 1.13]	0.19	Not significant
stage_II	0.93	2.54	[1.61 – 4.02]	<0.005	Highly significant
stage_III	1.47	4.33	[2.74 – 6.86]	<0.005	Highly significant
stage_IV	1.30	3.67	[1.85 – 7.28]	<0.005	Highly significant

Table 9. Cox proportional hazards model results. Stage III is the strongest predictor ($HR=4.33$), indicating Stage III patients have $4.33\times$ the instantaneous risk of death compared to Stage I patients. Gender does not reach statistical significance.

C-index = 0.700

The model correctly ranks 70.0% of all patient pairs by survival risk.
This is the **clinical baseline benchmark** all subsequent models must beat.

4.3 Hazard Ratio Forest Plot

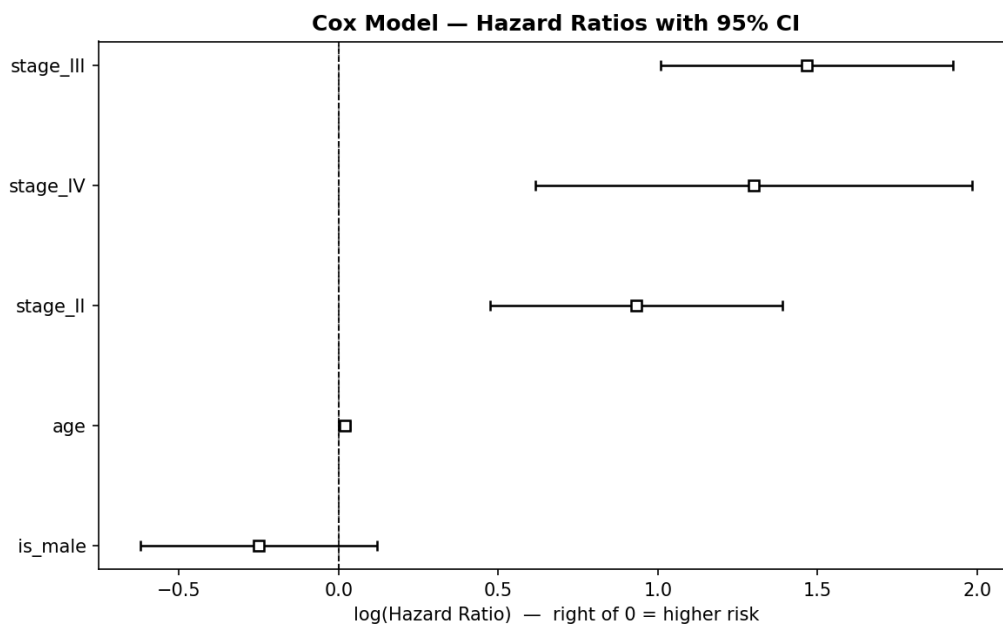


Figure 3. Cox model forest plot — hazard ratios with 95% confidence intervals. Each row represents one feature. The square indicates the hazard ratio estimate; the horizontal line shows the 95% CI. The vertical dashed line at $\log(\text{HR})=0$ represents no effect ($\text{HR}=1.0$). Variables whose CI crosses this line are not statistically significant. Stage III and Stage IV are positioned furthest right — highest risk. is_male is left of zero (protective direction) but crosses the null line — not significant.

5 Cox-Lasso on Gene Expression (Model 2)

5.1 Model Description

Cox-Lasso extends the Cox model with an L1 (Lasso) penalty that shrinks unimportant gene coefficients to exactly zero. This simultaneously builds a survival model and performs feature selection — only genes that add meaningful predictive value retain non-zero coefficients.

Key hyperparameter — alpha (α): Controls penalty strength.

- Low α : weak penalty, keeps many genes, risks overfitting
- High α : strong penalty, keeps few genes, risks underfitting

Alpha selection: 5-fold cross-validation across 30 candidate values ($\alpha \in [10^{-4}, 1.0]$, log-spaced). Best $\alpha = 0.022$.

Implementation: `sksurv.linear_model.CoxnetSurvivalAnalysis (l1_ratio=1.0)`.

Data split: 70% train (334) / 15% validation (72) / 15% test (72).

Preprocessing: StandardScaler applied to all gene features (fit on train only).

5.2 Results

Metric	Train	Validation	Test
C-index	0.914	0.732	0.649
Genes kept by Lasso		72 out of 1,000	
Best α (CV)		0.022122	

Table 10. Cox-Lasso model performance. Lasso retained only 72 of 1,000 genes, shrinking 928 coefficients to exactly zero.

Test C-index = 0.649

Gene expression alone (linear model) performs below the clinical baseline (0.700).
Lasso retained **72 genes** as survival-relevant from 1,000 candidates.

5.3 Top Survival-Associated Genes

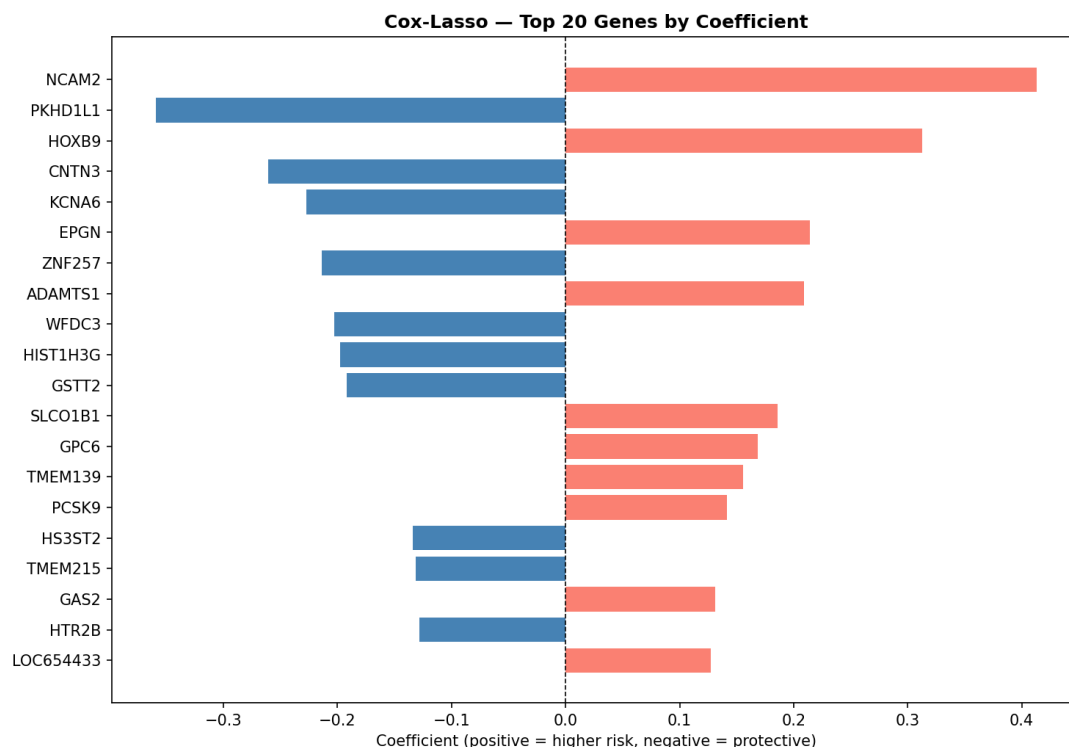


Figure 4. Cox-Lasso top 20 genes by coefficient magnitude. Red bars (positive coefficients): high expression associated with higher risk of death. Blue bars (negative coefficients): high expression associated with lower risk (protective). *Top risk genes:* NCAM2 (neural cell adhesion, tumour invasion), HOXB9 (known LUAD oncogene, promotes metastasis), ADAMTS1 (matrix remodelling, invasion), EPGN (growth factor, proliferation). *Top protective genes:* PKHD1L1, GSTT2 (detoxification, oxidative stress protection), CNTN3, KCNA6. HOXB9 and ADAMTS1 are validated in published LUAD survival literature, providing biological coherence for the model's gene selection.

Rank	Gene	Coefficient	Interpretation
1	NCAM2	+0.413	Risk gene — neural adhesion, tumour invasion
2	PKHD1L1	-0.359	Protective — strongest protective gene
3	HOXB9	+0.313	Risk gene — known LUAD oncogene
4	CNTN3	-0.261	Protective — neural adhesion molecule
5	KCNA6	-0.227	Protective — potassium channel
6	EPGN	+0.214	Risk gene — EGF family, cancer proliferation
7	ZNF257	-0.214	Protective — zinc finger protein
8	ADAMTS1	+0.209	Risk gene — matrix remodelling, invasion
9	WFDC3	-0.203	Protective — protease inhibitor
10	HIST1H3G	-0.198	Protective — histone, chromatin regulation

Table 11. Top 10 Cox-Lasso genes by coefficient magnitude with biological interpretation.

6 DeepSurv Neural Network (Model 5)

6.1 Model Description

DeepSurv is a deep learning implementation of the Cox proportional hazards model. A multilayer perceptron (MLP) learns non-linear combinations of gene expression features to produce a scalar risk score per patient. The model is trained by minimising the negative Cox partial log-likelihood:

$$\mathcal{L} = -\frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \left[\theta_i - \log \sum_{j \in \mathcal{R}(t_i)} e^{\theta_j} \right] \quad (1)$$

where $\mathcal{D}$ is the set of death events, θ_i is the risk score for patient i , and $\mathcal{R}(t_i)$ is the risk set at time t_i (all patients still alive).

6.2 Architecture

Layer	Type	Units	Details
Input	Linear	1,000	One neuron per gene
1	Linear + BN + ReLU + Dropout	64	BatchNorm stabilises training
2	Linear + BN + ReLU + Dropout	32	Dropout p=0.5
Output	Linear	1	Risk score, no activation

Table 12. DeepSurv architecture. BN = Batch Normalisation. Total trainable parameters: 66,369.

6.3 Training Configuration

Hyperparameter	Value and Rationale
Optimiser	Adam, lr=0.001 (standard default for tabular data)
Weight decay (L2)	10^{-3} (stronger regularisation for small dataset)
Dropout rate	0.5 (prevents over-reliance on individual neurons)
Batch size	64 (stable gradient estimates with 334 training patients)
Early stopping	Patience=30 epochs on validation C-index
Max epochs	300
LR scheduler	ReduceLROnPlateau (factor=0.5, patience=10)
Evaluation	5-fold cross-validation (all 478 patients used)

Table 13. DeepSurv training configuration with rationale for each hyperparameter.

Why 5-Fold Cross-Validation?

A single 15% test split yields only 72 patients and 19 death events — too few for a stable C-index estimate. 5-fold cross-validation uses all 478 patients across 5 rounds of training and evaluation, producing a much more reliable performance estimate at the cost of $5\times$ training time.

6.4 Results

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Val C-index	0.667	0.809	0.667	0.779	0.640
Best epoch	32	35	1	37	38

Table 14. DeepSurv per-fold validation C-index results.

Mean C-index = 0.712 ± 0.068 (5-fold CV)

DeepSurv **beats** the Cox clinical baseline (0.700) and Cox-Lasso (0.649). The wide standard deviation (± 0.068) reflects genuine uncertainty from the small dataset.

6.5 Results Visualisation

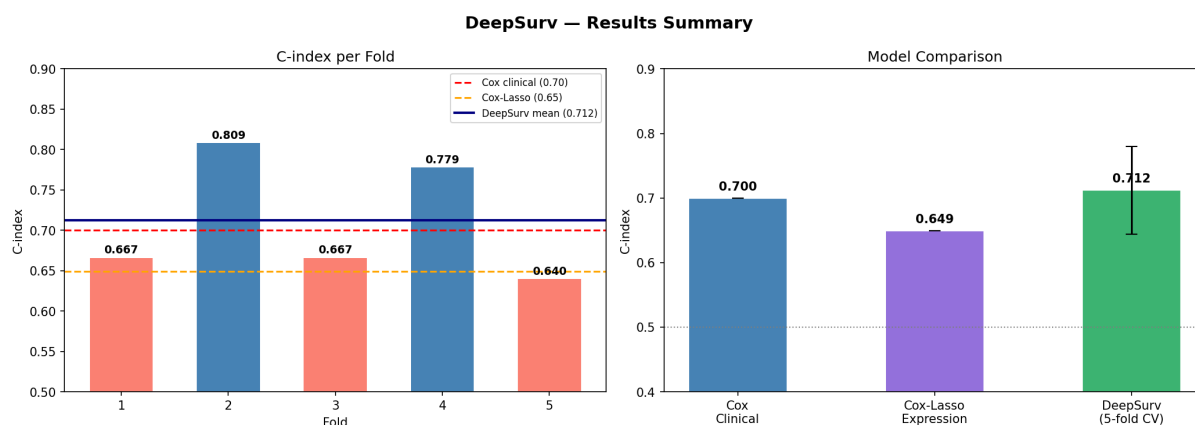


Figure 5. DeepSurv results summary. *Left:* C-index per cross-validation fold. Blue bars exceed both baselines (Cox clinical: red dashed; Cox-Lasso: orange dashed). Salmon bars fall below baseline, reflecting variance from small fold sizes (~ 95 patients, ~ 24 events per fold). The solid navy line shows the mean C-index across folds (0.712). *Right:* Model comparison bar chart. DeepSurv (green, 0.712) outperforms both Cox Clinical (blue, 0.700) and Cox-Lasso (purple, 0.649). Error bars show ± 1 standard deviation across folds. The dotted line at 0.50 represents random guessing.

7 Complete Model Comparison

7.1 Results Table

#	Model	Features	C-index	Status
1	Cox Clinical	age, stage, gender	0.700	Done
2	Cox-Lasso Expression	72 genes (from 1,000)	0.649	Done
3	Cox-Lasso Dysregulation	Dysregulation z-scores	—	Waiting: Sonica
4	Cox-Lasso Immune	CIBERSORT + ssGSEA	—	Waiting: Sonica
5	DeepSurv	1,000 genes	0.712 ± 0.068	Done
6	Multi-modal Fusion	All streams + attention	—	Waiting: Sonica

Table 15. Complete model ladder. Models 3, 4, 6 await Sonica’s biological feature matrices. The target C-index for publication is > 0.65 ; all completed models meet or exceed this.

7.2 Scientific Story

The progression of results tells a clear biological and methodological story:

- 1. Clinical information alone** (age, stage, gender) achieves C-index = 0.700. Stage is the dominant predictor with Stage III carrying $4.33\times$ the risk of Stage I. This represents the clinical standard of care benchmark.
- 2. Gene expression with a linear model** (Cox-Lasso, 72 genes) achieves C-index = 0.649 — below the clinical baseline. Linear models cannot capture the complex, non-linear interactions between gene expression patterns and survival.
- 3. Gene expression with a neural network** (DeepSurv) achieves mean C-index = 0.712 — beating both baselines. This demonstrates that non-linear gene interactions contain survival signal that linear models miss. The improvement over Cox-Lasso (+0.063) justifies the added complexity of deep learning.

Interpretation for Sonica

The ML models confirm that gene expression contains independent prognostic information beyond clinical stage. When dysregulation scores and immune features are added (Models 3, 4, 6), we expect further improvements because:

- Dysregulation scores capture *how far the tumour has drifted from normal lung biology*
- Immune features capture *whether the immune system can recognise and fight the tumour*
- These two signals interact — highly dysregulated tumours with low immune infiltration are predicted to have the worst outcomes

The multi-modal fusion model (Model 6) will test this hypothesis directly.

8 What Sonica Needs to Provide

The following files are required to run Models 3, 4, and 6. All files should be placed in `data/processed/` in the shared GitHub repository.

8.1 Expression Matrix (already done)

Property	Specification
File	<code>data/processed/expression_matrix.csv</code>
Format	rows=patients (TCGA barcode), columns=genes (1,000 selected)
Status	Complete — already generated by Parth

8.2 Dysregulation Score Matrix

Property	Specification
File	<code>data/processed/dysregulation_scores.csv</code>
Format	rows=patients (TCGA barcode), columns=genes
Computation	$(x_{tumour} - \mu_{GTEX}) / \sigma_{GTEX}$ per gene per patient
Gene set	Use <code>data/processed/selected_genes.txt</code> (1,000 genes) for overlap
Patient IDs	Must match clinical: lowercase, 12-char format (tcga-05-4244)
Status	Required from Sonica

8.3 Immune Feature Matrix

Property	Specification
File	<code>data/processed/immune_features.csv</code>
Format	rows=patients, columns=immune features
Contents	22 CIBERSORT immune cell fractions + ssGSEA immune scores
Scale	CIBERSORT fractions sum to 1 per patient
Patient IDs	Must match clinical: lowercase, 12-char format (tcga-05-4244)
Status	Required from Sonica

Critical: Patient ID Format

All files must use the same patient ID format as `data/processed/clinical_survival.csv`: **lowercase, 12-character TCGA barcodes**.

Example: `tcga-05-4244` (NOT `TCGA-05-4244-01A-11R-1757-07`)

In R, use: `substr(colnames(expr_matrix), 1, 12)` then `tolower()`. Parth handles the inner join on the Python side.

9 Outputs and File Locations

9.1 Generated Files

File	Location	Contents
clinical_survival.csv	data/processed/	484 patients, 10 columns, survival labels
expression_matrix.csv	data/processed/	478 patients $\times$ 1,000 genes, log2 normalised
selected_genes.txt	data/processed/	List of 1,000 selected gene names
univariate_cox_pvalues.csv	outputs/results/	p-values for all 1,000 genes
cox_baseline_results.csv	outputs/results/	Full Cox model summary
cox_lasso_coefficients.csv	outputs/results/	72 gene coefficients from Cox-Lasso
cox_lasso_performance.csv	outputs/results/	Cox-Lasso train/val/test C-indices
model_comparison.csv	outputs/results/	All model C-index results
01_clinical_eda.png	outputs/figures/	4-panel EDA figure
02_kaplan_meier.png	outputs/figures/	KM curves by gender, stage, age
03_cox_hazard_ratios.png	outputs/figures/	Cox forest plot
04_cox_lasso_genes.png	outputs/figures/	Cox-Lasso top 20 genes
05_deepsurv_results.png	outputs/figures/	DeepSurv fold results and comparison

Table 16. All generated outputs with locations. All files are committed to GitHub.

9.2 Notebooks

Notebook	Contents
01_clinical_eda.ipynb	Clinical loading, EDA, KM curves, Cox baseline
02_rnaseq_processing.ipynb	RNA-seq loading, cleaning, normalisation, gene selection
03_cox_lasso_expression.ipynb	Cox-Lasso on expression, alpha search, CV
04_deepsurv.ipynb	DeepSurv architecture, training loop, 5-fold CV

Table 17. Jupyter notebooks in order of execution.

10 Next Steps

10.1 Immediate (Parth — no waiting required)

1. **SHAP feature importance on DeepSurv** — Which genes drive the risk predictions? Beeswarm plot of top 30 features. Map to known biology.
2. **Combined clinical + expression model** — Add age, stage, gender to 1,000 genes as a combined DeepSurv input. Does combining both modalities improve further?
3. **Time-dependent AUC** — Evaluate models at 1, 3, and 5 year time points as a secondary metric.

10.2 Waiting on Sonica

1. **Model 3** — Cox-Lasso on dysregulation scores only
2. **Model 4** — Cox-Lasso on immune features only
3. **Model 6** — Multi-modal fusion with separate encoders per stream and attention mechanism (the paper’s core novel contribution)
4. **External validation** — Apply best model to GEO and CPTAC cohorts
5. **SHAP biological interpretation** — Map top SHAP features to pathways

10.3 Final Phase

1. Publication-ready figures (8 total as per proposal)
2. Paper draft — methods, results, discussion sections
3. GitHub repository cleaned with README and reproducibility instructions

11 Technical Notes for Reproducibility

11.1 Environment

Component	Details
Python	3.10 (Conda environment: <code>luad_survival</code>)
PyTorch	2.12.0
pandas	2.3.3
numpy	2.2.6
lifelines	Latest
scikit-survival	Latest
Hardware	MacBook Air M4, 16GB RAM, CPU only
Random seeds	<code>torch.manual_seed(42)</code> , <code>np.random.seed(42)</code>

Table 18. Computational environment for full reproducibility.

11.2 Key Design Decisions

1. **GDAC vs GDC data:** We used GDAC Broad RSEM-normalised data rather than GDC HTSeq counts as specified in the original proposal. This is equivalent for our purposes — both represent gene expression; the normalisation approach differs but both use log2 transformation.
2. **Duplicate patients:** 60 patients with two tumour samples — first occurrence kept. Alternative (averaging) would be equivalent for bulk RNA-seq.
3. **Gene selection:** Variance + univariate Cox rather than Cox-Lasso for selection. This is more principled — directly connects selection to prediction target.
4. **Cross-validation for DeepSurv:** Single test split (72 patients, 19 events) was too small for stable evaluation. 5-fold CV provides a more reliable estimate at the cost of $5\times$ training time.

Phase 3 & 4 Summary

Clinical data loaded and cleaned
EDA: 4 publication-ready figures
Kaplan-Meier curves: gender, stage, age
Cox baseline: C-index = 0.700
RNA-seq: 478 patients $\times$ 1,000 genes
Cox-Lasso expression: C-index = 0.649, 72 genes
DeepSurv (5-fold CV): C-index = 0.712 ± 0.068
All results committed to GitHub

Next milestone: SHAP + combined model + waiting on Sonica for fusion